

Final Assignment

I. Project requirements

1. Project Objective

The objective of this project is to design, implement, and document an end-to-end NLP system that solves a realistic industry or business problem.

This project must reflect real-world NLP practice, including problem formulation, data handling, model development, deployment considerations, and ethical responsibility.

Important: This is not a research-only or notebook-only project. Your system must be designed as if it were going into production.

2. Business Problem Definition

- Requirements: You must begin with a clear business use case where NLP provides measurable value.

- Deliverables: Submit a Problem Definition Document (1-2 pages) containing:

- Business context and motivation
- Target users or stakeholders
- Description of the problem being solved
- Explanation of why NLP is required
- Success metrics, including:
 - Business metrics (e.g., cost reduction, time saved, efficiency)
 - Technical metrics (e.g., accuracy, F1-score, latency)

3. Development Infrastructure & Tooling

- Requirements: Your project must follow professional software development practices, including:

- Programming language: Python
- Version control: Git
- Clear and modular project structure
- Dependency management (e.g., requirements.txt, pyproject.toml)

- Deliverables:

- A source repository containing:
 - src/ – source code
 - data/ – data or data scripts

- models/ – trained models or checkpoints
- configs/ – configuration files
- tests/ – tests (where applicable)
- A README explaining:
 - Environment setup
 - How to train the model
 - How to run inference
 - Basic logging or experiment tracking

4. Data Management

- Requirements: You must demonstrate responsible and structured data handling:

- Data sourcing (public, synthetic, or anonymized)
- Data preprocessing and cleaning
- Train/validation/test split
- Handling missing, noisy, or biased data

- Deliverables: Submit a Data Description Document including:

- Data source and licensing (if applicable)
- Dataset size and language(s)
- Preprocessing steps
- Justification for data splits
- Discussion of known limitations and potential biases

5. Model Selection & Optimization

- Requirements:

- Select an appropriate NLP model (traditional ML or deep learning)
- Justify why the model fits the business problem
- Implement at least basic hyperparameter tuning
- Compare performance against at least one baseline

- Deliverables:

- Model architecture description
- Training procedure
- Evaluation results including:
 - Baseline comparison
 - Error analysis (what the model fails on and why)

- Discussion of trade-offs:
 - Accuracy vs. speed
 - Model complexity vs. Maintainability

6. Deployment

- Requirements: Your system must be deployable, not only trainable. Acceptable deployment formats include:

- REST API
- Web-based demo
- Command-line interface
- Batch processing pipeline
- You must consider:
 - Latency
 - Scalability
 - Model versioning

- Deliverables:

- A working inference pipeline
- Documentation describing:
 - User interaction
 - Input and output formats
- Discussion of deployment challenges and limitations

7. Agentic AI Component

- Requirements: Your project must include agentic behavior, such as:

- Multi-step reasoning
- Tool usage (e.g., search, APIs, databases)
- Decision-making based on intermediate outputs

- Examples include:

- A chatbot that retrieves documents before responding
- A system that routes inputs to different NLP models
- An application that asks clarifying questions

- Deliverables:

- Description of the agent architecture
- Flow diagram or pseudocode
- Example interaction demonstrating agent decisions

8. Continual Learning & Monitoring

- Requirements: You must design (conceptually) for model evolution over time, even if full implementation is not feasible.

- Deliverables:

- Continual learning strategy document describing:
 - How new data would be collected
 - How retraining or fine-tuning would occur
 - How performance degradation would be detected
- Proposed monitoring metrics
- Discussion of model drift risks and mitigation strategies

9. Data Privacy & Model Robustness

- Requirements: Your system must address:

- Data privacy and security risks
- Robustness to noisy or adversarial inputs

- Deliverables:

- Privacy analysis covering:
 - Handling of personally identifiable information (PII)
 - Anonymization or minimization strategies
- Robustness discussion including:
 - Out-of-domain inputs
 - Failure cases
- Mitigation strategies

10. Project Management & Teamwork

- Requirements:

- You must demonstrate team-style planning.

- Deliverables:

- Project plan or timeline

- Task breakdown (roles may be simulated for solo work)
- Reflection on how the project would scale in a real team environment

11. Ethics & Responsible AI

- Requirements:

- You must critically assess the ethical implications of your system.

- Deliverables:

- Ethics impact statement discussing:
 - Who benefits from the system
 - Who could be harmed
- Bias and fairness risks
- Explainability considerations for non-technical stakeholders
- Discussion of potential misuse

II. Submission requirement

Students must submit three components:

- a written report,
- the complete source code, and
- a presentation slide deck.

Note: Failure to submit any required component may result in a significant grade penalty.

1. For the written report

- Format:

- Length: 10–15 pages (excluding references and appendices)
- File type: PDF
- Font size: 11–12 pt
- Line spacing: 1.15–1.5
- Figures and tables must be clearly labeled and referenced

- Content: The report must contain all information in the Section I.

2. Source code submission

- Format:

- Submission via Git repository link (public repo)
- Repository must be complete and runnable

- Required Repository Structure: At minimum, the repository must contain:

```
project-root/
├── src/           # Core source code
├── data/          # Data or data-loading scripts
├── models/        # Trained models or checkpoints
├── configs/       # Configuration files
├── tests/         # Tests (if applicable)
├── requirements.txt or pyproject.toml
└── README.md
```

- The README.md must include:

- Project overview
- Environment setup instructions
- Dependency installation steps
- How to train the model
- How to run inference on the deployed system
- Description of deployment method

- Code Quality Expectations:

- Code must be:
 - Well-organized and modular
 - Clearly commented where necessary
 - Reproducible (docker is encouraged)
- Hard-coded paths, credentials, or secrets are not allowed
- Large datasets should not be committed directly (use download scripts)

3. Presentation slides

- Format:

- Length: 10-15 slides
- File type: PDF or PPTX

- Slides must include:

- Title & Team Information
- Business Problem & Motivation
- Proposed NLP Solution
- System Architecture Diagram
- Data Overview
- Model & Evaluation Results
- Agentic AI Component

- Deployment Overview
- Ethics, Privacy & Risks
- Key Takeaways & Future Work

- Presentation Expectations:

- Slides should be visual and concise
- Avoid dense text blocks
- Diagrams and charts are strongly encouraged
- Slides must align with the written report

II. Project list

1. Sign Language Translation System

Link: <https://www.wmt-slt.com/>

2. Resume Screening & Ranking System

Link: <https://github.com/Hunterdii/Smart-AI-Resume-Analyzer>

3. Knowledge Base Question-Answering System

Link: <https://github.com/LHRLAB/ChatKBQA>

4. Invoice & Receipt Processing System

Link: <https://github.com/ruizguille/invoice-processing>

5. Audiobook Generation System

Link: <https://github.com/denizsafak/abogen>

6. Vietnamese Handwriting Detection System

Link: To be defined by the student

7. Document-Level OCR System

Link: To be defined by the student

8. Exploratory Search of Scientific Literature System

Link: <https://github.com/NLP-Knowledge-Graph/NLP-KG-WebApp>

9. Citation Recommendation System

Link: <https://github.com/nianlonggu/SciLit>

10. Automatic Meeting Minutes & Action Item Extraction System

Link: <https://github.com/Zackriya-Solutions/meeting-minutes>

11. Fake News & Misinformation Detection System

Link: <https://github.com/KaiDMML/FakeNewsNet>

12. Model Drift Detection & Monitoring System for NLP

Link: <https://github.com/SeldonIO/alibi-detect>

13. Speech-to-Speech Translation System

Link: https://github.com/facebookresearch/seamless_communication

14. Structured Document-Level Translation System

Link: <https://aclanthology.org/2025.ijcnlp-long.38.pdf>

15. Document Image Machine Translation System

Link: <https://aclanthology.org/2025.acl-long.606.pdf>

16. Hate Speech & Toxicity Detection System

Link: <https://github.com/unitaryai/detoxify>

17. Multimodal Question-Answering System

Link: <https://github.com/facebookresearch/mmf>

18. Temporal Knowledge Graph Question-Answering System

Link: <https://github.com/cosmicexotic/TKGQA-Survey>

19. Text-to-Image Retrieval System

Link: <https://github.com/kingyiusuen/clip-image-search>

20. Text Search within Images System

Link: <https://github.com/kanchan2803/ImgToText>

21. Others

This is the project that your team proposes (it needs to be approved).